

A world map with a dark blue background. The landmasses are colored in various shades of orange, red, and green, representing different COVID-19 data points. The colors are more vibrant in some regions, like North America and Europe, and more muted in others, like South America and Africa. The map is centered on the Atlantic Ocean.

AN INTERACTIVE COVID-19 DECISION SUPPORT APPLICATION

Integrating Machine Learning, Clustering, and Explainable AI

Daryl Morris

Springboard Data Science Career Track
Capstone Three

July 2026

PROJECT EVOLUTION



Capstone Two

Predictive Machine Learning Model

- Random Forest regression model
- Country-level COVID-19 mortality prediction
- Static analysis and feature importance
- Machine learning performance evaluation

Evolved Into



Capstone Three

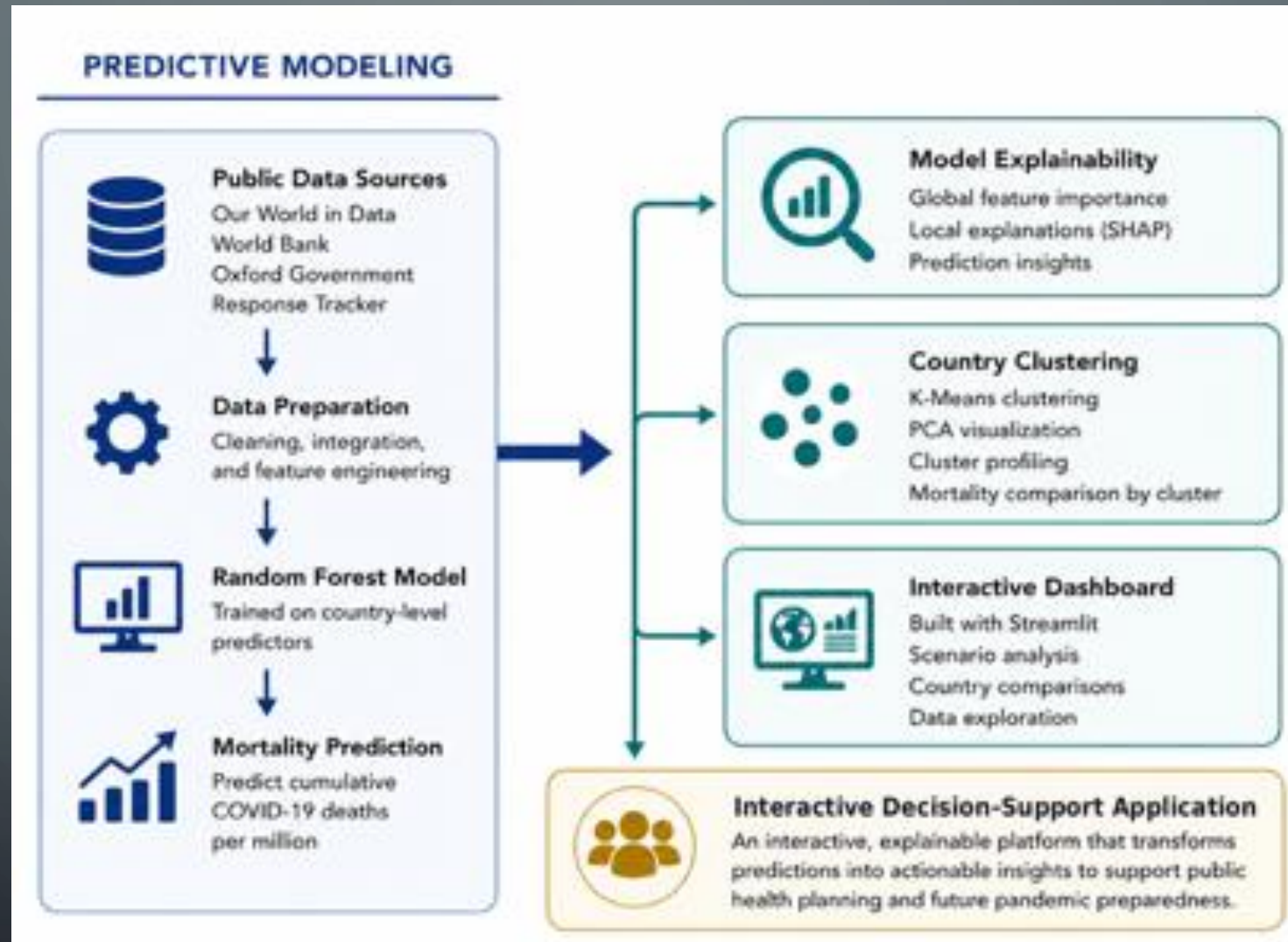
Interactive Decision-Support Application

- Interactive Streamlit dashboard
- K-Means country clustering
- Scenario-based prediction
- SHAP explainability
- Real-time prediction explanation

Project Goal

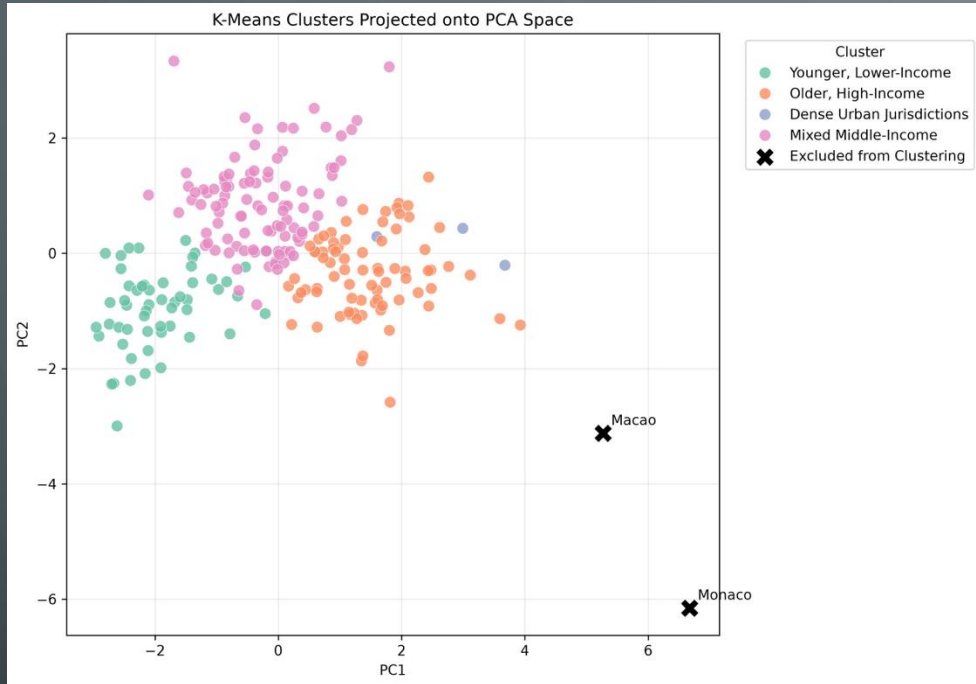
Transform predictive machine learning into an interactive, explainable decision-support application.

SYSTEM ARCHITECTURE

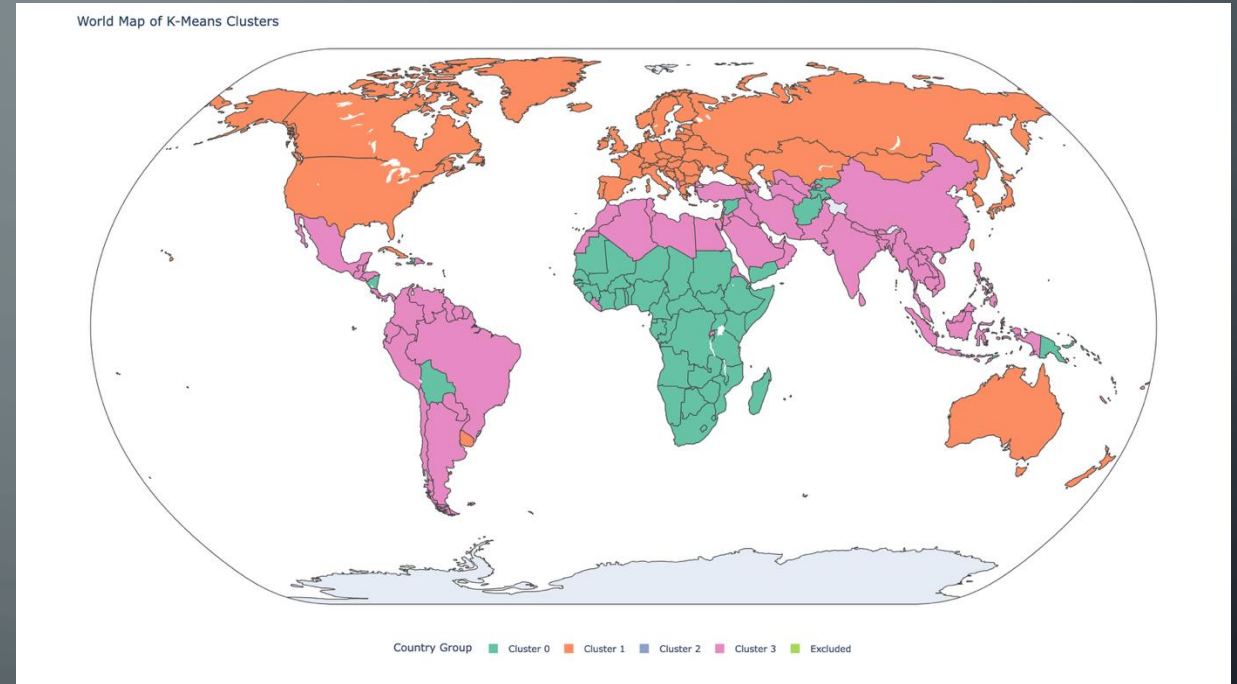


Integrates machine learning prediction, country clustering, and SHAP explainability into a unified interactive application.

COUNTRY EXPLORATION THROUGH CLUSTERING



K-Means Cluster Visualization (PCA Projection)

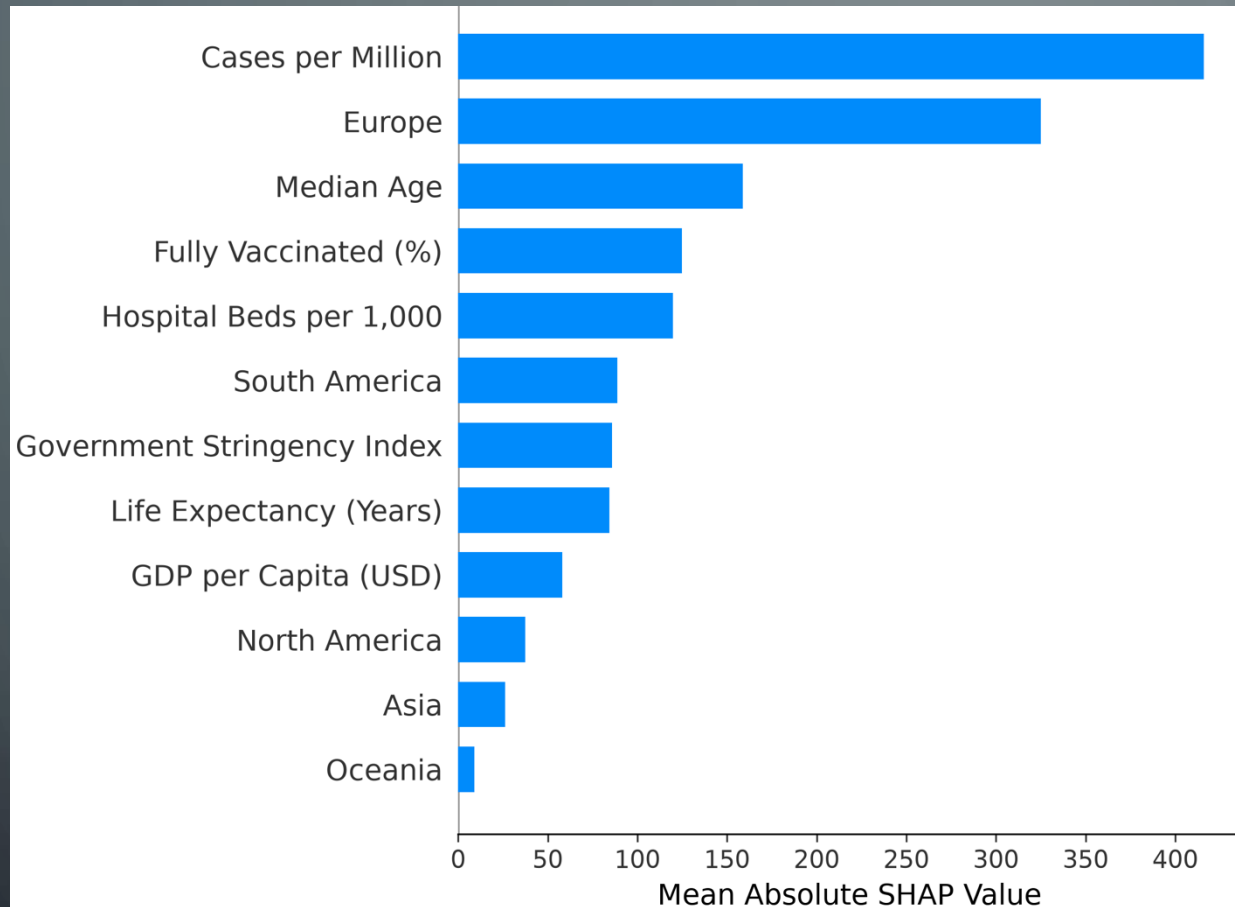


Interactive Country Cluster Map

Key Insight

Countries are grouped according to demographic, healthcare, economic, and public health characteristics. COVID-19 mortality was intentionally excluded from the clustering process.

EXPLAINABLE AI USING SHAP



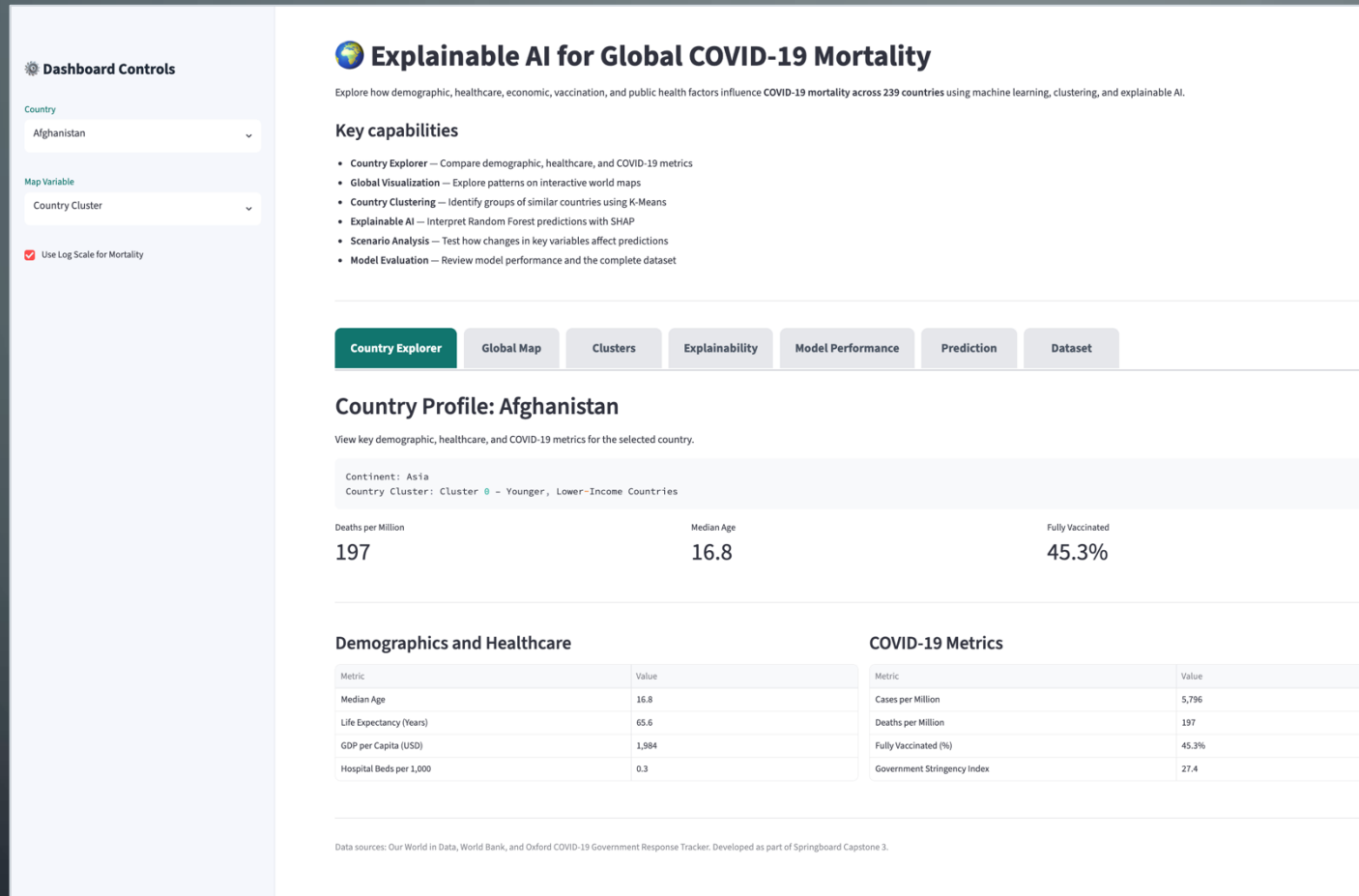
What SHAP Explains

- Explains individual predictions
- Quantifies feature contributions
- Identifies the most influential predictors
- Opens the Random Forest “black box”

Key Insight

Cases per Million and geographic region are the model's strongest predictors of COVID-19 mortality. SHAP provides transparent explanations of how these and other variables influence model predictions.

INTERACTIVE DASHBOARD



Dashboard Modules

- Country Explorer
- Global Map
- Country Clusters
- SHAP Explainability
- Model Performance
- Scenario Prediction
- Dataset Browser

Key Insight

The dashboard integrates multiple analytical tools into a single interface, allowing users to move seamlessly between exploration, prediction, clustering, and explainable AI.

INTERACTIVE SCENARIO ANALYSIS

Changed Inputs (Baseline: Cuba)

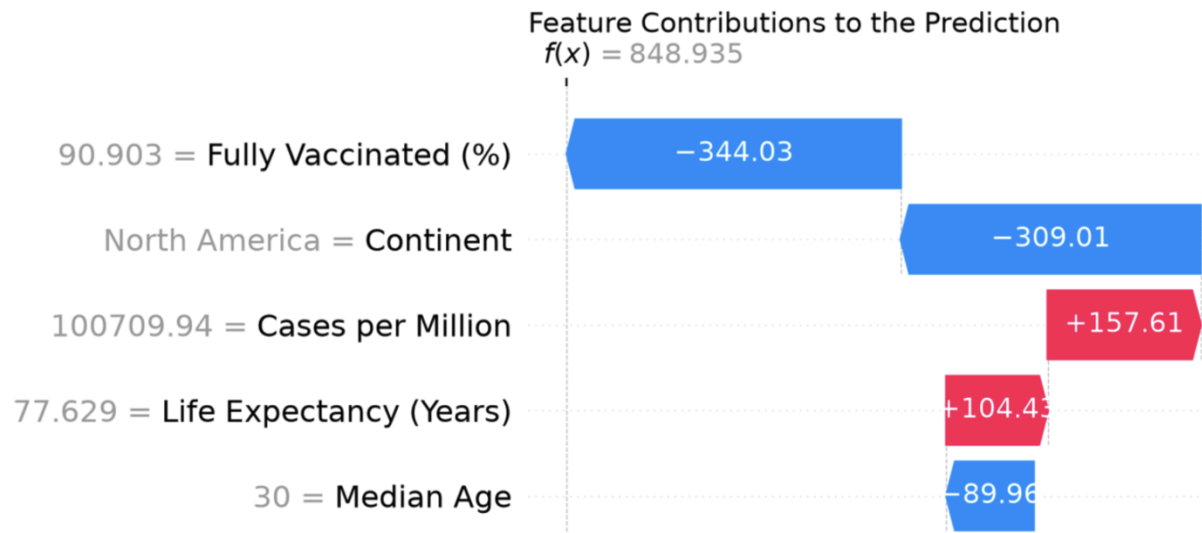
Input	Original	Modified
Median Age	41.5	30.0 (-11.5)

Prediction Explanation

The chart below explains how each feature influenced the predicted COVID-19 mortality. Positive values increase the prediction, while negative values decrease it.

▲ Increased Prediction: Cases per Million (+158), Life Expectancy (Years) (+104), Hospital Beds per 1,000 (+39)

▼ Decreased Prediction: Fully Vaccinated (%) (-344), Continent (-309), Median Age (-90)



What Users Can Do

- Select any country
- Modify demographic, healthcare, economic, or policy variables
- Generate a new mortality prediction instantly
- Understand direct effects and model interactions
- Explain every prediction using SHAP

Key Insight

Users can test hypothetical scenarios and immediately understand not only how the prediction changes, but which variables are responsible for that change.

PROJECT DELIVERABLES

Data Engineering

- Integrated public datasets
- Engineered country-level features

Machine Learning

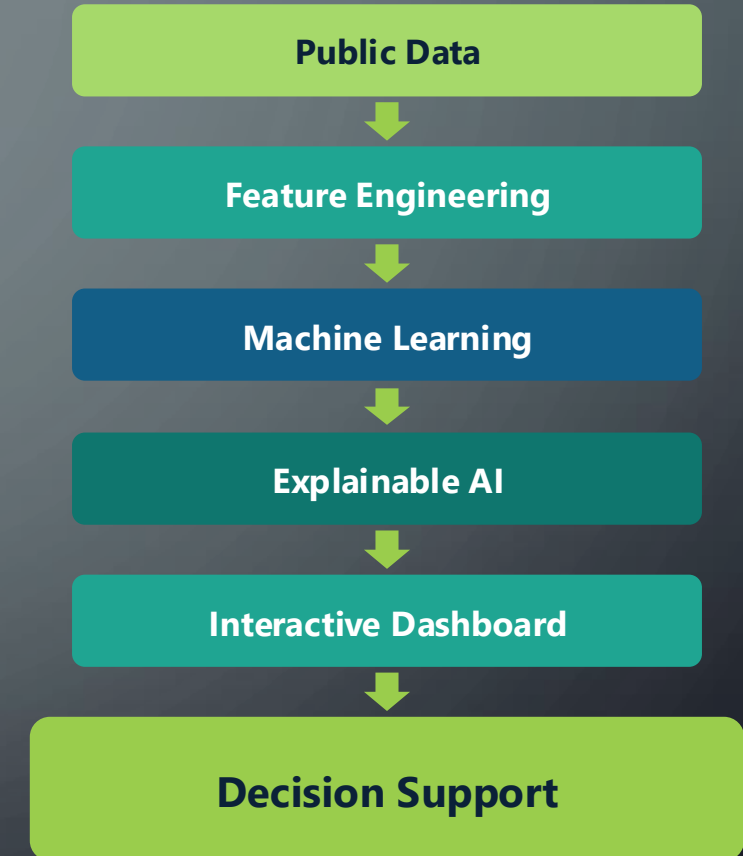
- Random Forest mortality prediction
- K-Means country clustering

Explainable AI

- Global and local SHAP explanations
- Feature interaction analysis

Interactive Dashboard

- Streamlit application
- Seven integrated analytical modules



Key Insight

This project extends beyond predictive modeling by combining data engineering, machine learning, clustering, explainable AI, and interactive visualization into a unified decision-support system for public health analysis.

CONCLUSIONS & FUTURE WORK

Conclusions

- ✓ Integrated multiple public datasets into a unified analytical dataset
- ✓ Developed an accurate Random Forest model for COVID-19 mortality prediction
- ✓ Applied SHAP to provide transparent global and local model explanations
- ✓ Built an interactive decision-support dashboard for country-level analysis

Future Work

- Update with newer COVID-19 datasets
- Extend the framework to temporal forecasting
- Evaluate additional machine learning models
- Deploy the dashboard as a public web application

Final Takeaway

This project demonstrates that integrating machine learning, explainable AI, and interactive visualization can create practical decision-support tools for understanding complex public health outcomes.